# Impulse changes momentum

**antocciphysics**

UNIT 3 / DAY 3 / HONORS

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect force over time to momentum change.

**SUCCESS CRITERIA - I can...**

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

An ideal circular orbit has GM=**225 m³/s²** and r=**9 m**. **Find speed**, inward acceleration and period; **explain how velocity changes**.

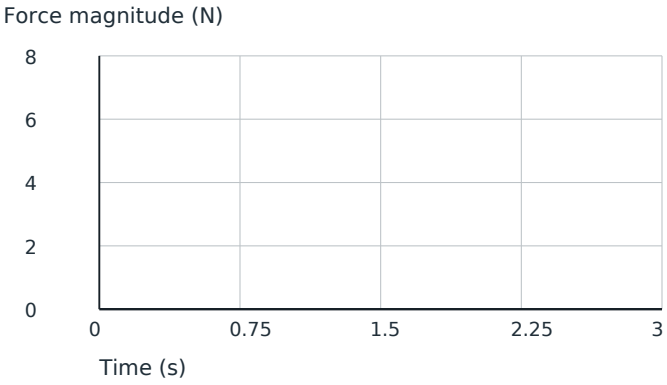
**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; $F_{avg}$ is average net force. Signed force-time area gives impulse; i = initial, f = final.

**Worked reasoning:** A constant **6 N** force acts right for **2 s**. Impulse is **12 N·s** right, so momentum increases by **12 kg·m/s**.

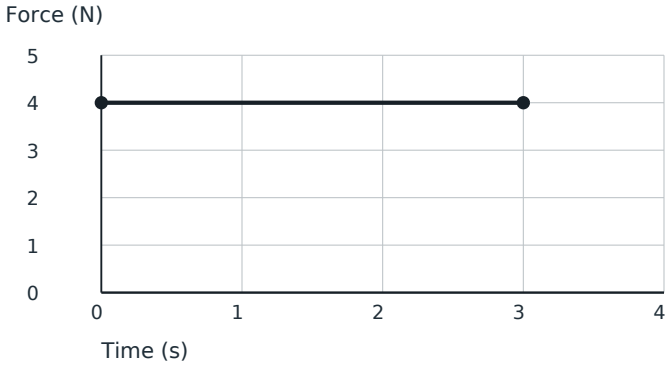
**Your turn:** 1. **Draw the force-time rectangle** and **shade its area**.
2. **Explain why the two impulse** units are equivalent.




**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

1. **Read the rectangular force-time graph**.
2. **Find the area** and **state what it means**.




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A **0.50 kg** ball hits a wall while moving at **+8 m/s** and rebounds at **-6 m/s**. Contact lasts **0.020 s**.
1. **Label the before-and-after diagram**.
2. **Calculate impulse** and average net force on the ball.

| **Before** | **After** |
| --- | --- |
| | |

Label mass and velocity.

**Marzano check - circle:** ① ② ③

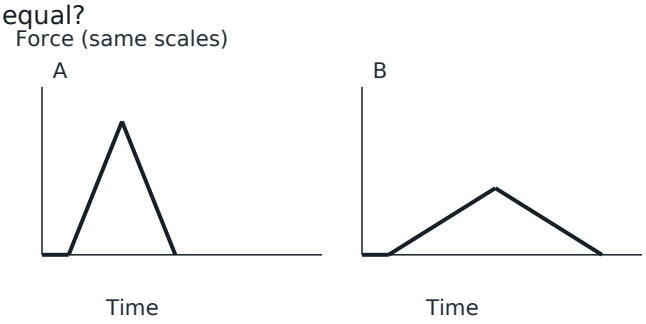
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## 05 Explain your reasoning

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

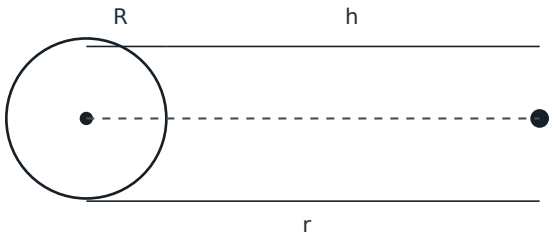



Same scales: compare impulse and peak force.

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

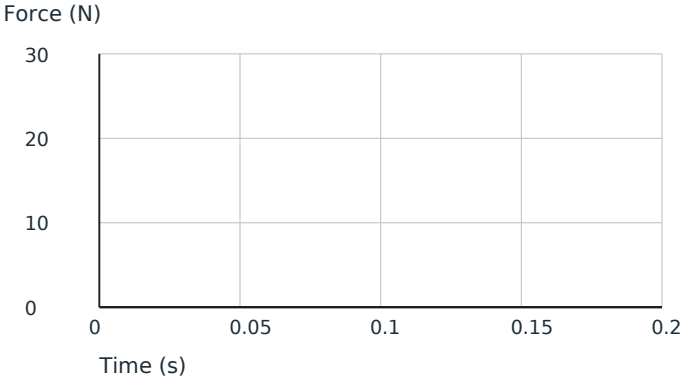
At what altitude h/R is a spherical planet's field **1/16** of its surface value? Repeat for **1/36**.
**Label the masses**, field or force relevant to the question.




**Marzano check - circle:** 1 2 3

## 07 Exit check

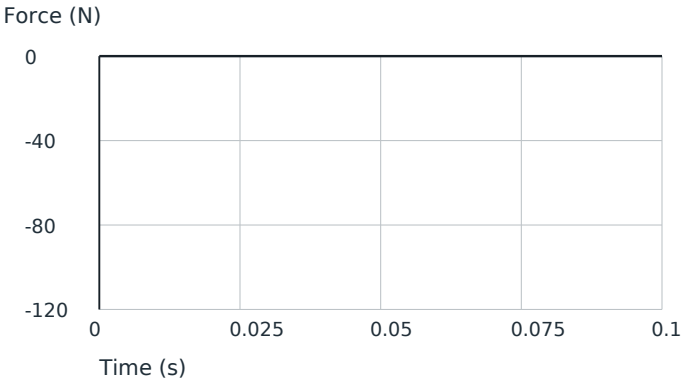
A triangular force pulse has base **0.20 s** and peak **30 N**. **Find impulse** and velocity change for a **0.60 kg** object.




**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

A **2 kg** cart changes velocity from **+3** to **-2 m/s** in **0.1 s**.
1. **Find impulse** and average force.
2. **Sketch a force-time rectangle** with the correct signed area.




**Marzano check - circle:** 1 2 3